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PREVIEW

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A study of the nature of objectively real time

Spitzer, Robert Joseph, Ph.D.

The Catholic University of America, 1989

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PREVIEW

THE CATHOLIC UNIVERSITY OF AMERICA

A Study of the Nature of Objectively Real Time

A DISSERTATION

Submitted to the Faculty of the

School of Philosophy

Of the Catholic University of America

In Partial Fulfillment of the Requirements

For the Degree of

Doctor of Philosophy



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Robert Joseph Spitzer, S.J.

Washington D.C.

1989

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This dissertation was approved by Paul Weiss, Ph.D., as director, and by William Wallace, Ph.D., and George McLean, Ph.D., as readers.

Paul Weiss
Director

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Preface

This study will attempt to elucidate and justify a model of objectively real time. Five interrelated, subsidiary objectives are involved: 1) to ontologically ground and justify the everyday experience of a) the past-present distinction as manifest in the distinction between ceased and existing occurrences, and b) earlier-later as manifest in faster and slower motions over equivalent distances, 2) to respond to Zeno's arguments against the reality of motion without denying the reality of time or continuous motion, 3) to reconcile the Aristotelian (Substance) and Bergsonian (Protomentalist) interpretations of the nature and ontological status of earlier-later, 4) to show the objective reality of an interval of the present, and 5) to mediate the data of past-present and earlier-later given in everyday experience with the data of variable length and time measurement given in the Special Theory of Relativity.

The study will consist of an introduction and three Chapters. The introduction will 1) locate this study's approach to objective time among other classical, modern, and contemporary studies, 2) summarize the data that will be used to ground the notion of objective time, 3) outline the major steps of the development of the notion of objective time, 4) discuss the various classical and modern positions

which are reconciled through this study's notion of objective time, and 5) assess J.E. McTaggart's argument against the reality of time.

The first chapter will be concerned with objectively real endurance. It will 1) isolate and ground the various objective, constituent principles of endurance, 2) show that there must be a minimum interval of objective endurance, 3) show the effects of such a minimum interval of endurance on projectile motion, and 4) show the superiority of a succession model of objective time over atomistic or flow models.

The second chapter will be concerned with the resultant model of time. It will 1) integrate a) the succession of minimum intervals of endurance, b) earlier-later, and c) past-present, 2) elucidate the consequent notion of event, 3) reveal the nature of temporal direction, and show its inviolable asymmetry, and 4) manifest the consequent ontological status of age and future.

The third chapter will 1) summarize the study's explanation of objective time, 2) systematize the metaphysical principles underlying this explanation, and 3) contrast this explanation with other classical and contemporary ones.

Introduction

This study is concerned with the nature of objectively real time. "Objectively real" means "existence independent of a human subject's cognitional activity." "Subjective" will refer to dependence on a subject's cognitional activity. Since objective time is entangled in many paradoxes, several contemporary philosophers and physicists doubt its existence.¹

I. The data of everyday experience:

A denial of objectively real time, however, must be reconciled with two data which periodically surface and reclaim the attention of philosophers: 1) the seeming objectively real distinction between past, present, and future occurrences, and 2) the non-spatial element in velocity. The first often comes to the attention of metaphysicians, and the second to physicists.

The first datum is an integral part of everyday experience. One is aware that certain occurrences no longer exist, and that the non-existence of these occurrences takes

¹G.J. Whitrow has summed up the controversy well: "The history of natural philosophy is characterized by the interplay of two rival philosophies of time--one aiming at its 'elimination' and the other based on the belief that it is fundamental and irreducible" (1980, 370).

The various positions of the major schools of thought on objectively real time will be presented in the footnotes throughout Chapters One and Two.

place in a sequential order. The perceived radical distinction between these non-existing occurrences and existing occurrences grounds the common sense distinction between past and present. Such is also the case with the distinction between existing occurrences (present) and occurrences which do not yet exist (future).

This common sense affirmation of the distinction between the various existential modes of occurrences manifests itself as objective (independent of human subjectivity) not only because there is common agreement about which occurrences no longer exist and which ones still do, but also because those which are thought no longer to exist do not exert themselves upon the experiencing subject in the same way as the ones which are thought to exist.

This common sense conviction of the objectivity of the distinction between past, present, and future is particularly tenacious,² because its denial implicitly requires denying the distinction between non-existing and existing occurrences.

Unfortunately, this distinction is often interpreted as a flow of events from the present to the past, or from the

²Einstein, who, as a physicist, was particularly critical of an objectivist view of time throughout most of his life (see Einstein 1964, 103-105; Hoffman 1972, 258; and Popper 1982, 89-92) told Carnap toward the end of his life that "there is something about the present Now that is essentially different from the past and the future, but that this is a difference which does not and cannot occur within physics" (Griffin 1986, 29 reporting on Carnap, 1963, 37f).

future to the past.³ This interpretation, which does not follow from the affirmation of a distinction between past, present, and future, gives rise to two paradoxes: 1) it implicitly attributes objective ontological status to past and future events and/or states, and 2) it spatializes temporal states (the future is a "place" where events flow from, and the past is a "place" where events flow to). These paradoxes, in turn, have led to the rejection of the objective reality of the distinction between past, present, and future. One aspect of this study will be concerned with showing that such an interpretation does not follow from the distinction between existing and non-existing occurrences.

A second datum grounding objective time manifests itself in degrees of velocity. Two elements are ingredient to projectile motion: 1) the distance traversed by the projectile, and 2) the speed with which it traverses this distance. Both of these elements appear to be objective; there is also general agreement about them in everyday experience.⁴

The objectivity of distance is easier to recognize than that of the speed with which the distance is traversed. The experience of the speed with which distance is traversed is

³For references, see the criticism of the flow model of time in Sections V and VII.

⁴The agreement is on the macroscopic level of experience. The relativity of length and duration of high velocity particles, etc. will be taken up in Chapter One, Sec IV.

fleeting in comparison with distance itself. Distance remains after the traversing has ceased. This tempts some physicists to interpret motion spatially.⁵

Distance cannot be the sole objective ground for the differentiation of locomotions. Without "earlier and later" such differentiation would be impossible.⁶ Suppose there are two projectiles traversing the same distance, and beginning together. If projectile #1 is faster than projectile #2, it will complete the trip earlier than projectile #2. If the distinction between earlier and later has no objective significance, neither does projectile #1's earlier arrival. And if projectile #1's earlier arrival has no objective significance, neither does its being faster than projectile #2.⁷

The objective significance of the distinction between earlier and later begins to fade when it is scrutinized

⁵As Einstein noticed "it is a characteristic of thought in physics...that it endeavors in principle to make do with "space-like" concepts alone" (1954, 141).

⁶Emile Meyerson strongly affirmed this in his effort to respond to the problem of the spatialization of time: "[The] transformation of time into space...exceeds the authority of the most clearly established facts and the most basic foundations of science" (1976, 356).

⁷The same holds true for asymmetrical processes. It makes no sense to attribute objective significance to the beginning and ending states of a process without attributing objective significance to earlier and later. Otherwise, the beginning and ending states would be coincident in time.

through the categories of physics.⁸ It eludes the classification of substance, energy, field, state, etc., and, most importantly, it lacks the seeming permanence of distances. A set of more fundamental ontological categories is needed to penetrate the nature and ground of objective fast and slow traversal, and objective earlier and later. The lack of such categories has led some philosophers and physicists to deny the objective reality of temporal distinctions altogether.⁹ Others attribute objective status to a static earlier-later which specifically excludes the present.¹⁰

⁸Although physics has very precise definitions of earlier-later as a measurement, it is prevented from defining the nature and ontological status of "objective, measurable earlier-later" (i.e. earlier-later considered as an objectively real measurable ground of temporal measurement), because the categories it uses to analyze the nature and ontological status of objective reality (e.g. energy, field, state) tend to be less fundamental than objective, measurable earlier-later. Henceforth, this study will be concerned with measurable earlier-later, rather than with earlier-later as a measurement.

⁹Louis De Broglie's interpretation of the theory of relativity is particularly bold: "In space-time, everything which for each of us constitutes the past, the present, and the future, is given en bloc..." (1949, 113).

Costa De Beauregard followed his mentor in his 1966 position, which he has since modified (as implied in 1987, 129-162): "There can no longer be any objective and essential...division of space-time into 'events which have already occurred' and 'events which have not yet occurred'" (1966, 429).

¹⁰Grünbaum, following the lead of Hermann Weyl, notes: "Let us now call the direction of entropy increase of a typical representative of these epochs the direction of 'later'...Our characterization of the difference between the two directions of time does not as such affirm the existence of a transient threefold division of events..." (Grünbaum

Perhaps the most common problem arising out of the attempt to interpret the second datum of objective time (i.e. the non-spatial element in velocity) without a set of fundamental ontological categories is the reduction of the threefold characterization of time (past-present-future) to a twofold characterization (earlier-later). A physicist examining differences in velocities or the asymmetry of physical processes may be inclined to admit that earlier-later has objective significance, but his data offers no compelling reason to admit to an objectively significant present.¹¹ Hence, many physicists and philosophers have denied the objectivity of the present, the distinction between past, present, and future, and becoming.¹² Not surprisingly, this tendency had its origin in the first philosopher to derive the definition of time from that of motion: Aristotle.¹³

1964, 414-417). These opinions will be examined in detail below and in the next chapter.

¹¹see Note #10 above.

¹²See for example Weyl and Grünbaum in Note #10.

¹³Aristotle made a radical departure from Plato's metaphysical approach to time. In the *Timaeus* (37D), Plato defined time to be the moving image of eternity (being), and, therefore the mediating principle between being and becoming: "[The creator] made this image eternal but moving according to number, while eternity rests in unity, and this image we call time. For there were no days and nights and months and years before the heaven was created, but when he constructed the heaven, he created them also. They are all parts of time, and past and future are created species of time which we unconsciously but wrongly transfer to eternal being, for we say that it 'was' or 'is' or 'will be,' but

the truth is that 'is' alone is properly attributed to it..." (Hamilton and Cairns, eds. Jowett, trans. 1961, 1167) See also Timaeus 32-34, 38A, 40D. See also the interpretations of these texts in Brumbaugh 1984, 27-44; Callahan 1948, 32-35; Cornford 1952, 96-104. These references refer only to Plato's later cosmology. Earlier formulations of the notion of time in Plato are analyzed in Brumbaugh 1984, 27-34 and Callahan 96-99.

Aristotle's definition shifted the focus from eternity/being to motion/change. In D,11 (219 b 2-3) of the Physics, he defines time in terms of a two-fold structure: "For time is just this: the number [arithmos in the sense of objective measurability rather than a subjective standard of measurement] of a motion with respect to the prior and the posterior" (Apostle, trans. 1980, 81). "Prior-posterior" here does not refer to the spatial aspect of motion (distance traversed). It is the non-spatial aspect intrinsic to the speed of the distance traversed. If there were no objective prior-posterior (earlier-later) fast or slow traversal of distance would not be distinguishable. Prior and posterior are just as integral to motion as here and there. Both are intrinsically measurable. By linking time to the non-spatial aspect of motion, Aristotle avoided the problem of spatializing time.

In addition to the time of motion, Aristotle linked time to the endurance of contingent existents ("the measure of their substance"). In D,12 he notes: "things which are destructible and generable, and, in general, things, which at one time exist, and at another do not, all of them must be in time; for there exists a greater time which will surpass their existence and the measure of their substance."--221 b 30-31--ibid p 85). This link of time to the endurance of contingent existents leads to the discussion of past-present-future in D, 13 : "...the present moment is both an end and a beginning of time, though not of the same time, but the end of what has passed and the beginning of the future...The term 'presently' signifies a part of future time which is near the present and indivisible moment, and 'already' signifies a part of past time not far from that moment" (222 b 1-3 & 8-10, ibid 86). This move represents an advance and a problem. The advance consists in moving the notion of time beyond locomotions (and other changes in spatial relations) to generation and corruption (being and nonbeing), and in implicitly linking this second aspect of time (being/nonbeing) to past-present-future. Unfortunately, he did not distinguish the two-fold structure of time associated with motion (prior-posterior) from the threefold structure of time associated with being/nonbeing (past-present-future). Instead, he